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Chapter 3

MARINE BIOFOULING IN HEAT EXCHANGERS

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ABSTRACT

It has been trying to summarize the knowledge obtained by our research team in the field of biofouling related with the power stations cooling water systems. A portable pilot plant has been used in order to carry out in-situ experimental studies as it simulates the condenser performance and also monitorize and characterize the biofouling development in different operation conditions. Two important energy industries have taken part in the studies allowing to set up the pilot plant in their facilities of Algeciras Bay, Spain. Chemical and biological methods have been used for the characterization of the biofouling. With the results obtained, it has been possible to improve the knowledge and modelization of this complex phenomenon in order to its mitigation.

1. THE PROBLEM OF FOULING IN SEAWATER SYSTEMS

1.1. Cooling Water Systems in Coastal Facilities

Seawater is employed aboard ships for refrigerating purposes in many heat exchangers-condensers of the main propulsion system as well as in the auxiliary equipment.

For similar purposes, and using the same cooling fluid, in cooling water systems in process industry and in conventional, combined cycle and nuclear power plants, huge amounts of sea water are employed in the condensers, where the unused energy is rejected as the steam is condensed after leaving the turbines providing the minimum back-pressure in order to achieve the maximum thermal efficiency in the thermodynamic power cycle.

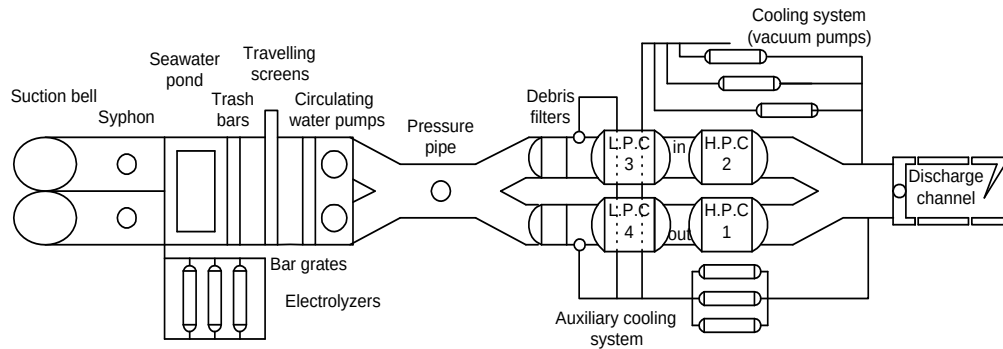


Figure 1.1. Once-through cooling system of a power station.

Surface condensers are basically *shell and tube* heat exchangers with the exhaust steam condensing on the shell side and the cooling water flowing in one or various passes inside the tubes. While small condensers are frequently cylindrical, larger power condensers are generally rectangular in shape.

The core of the equipment consists of the tube bundles with tens of thousands of tubes which receive and discharge the water from and to the water boxes or plenums. Cooling water velocities are usually dependent on a number of factors including economics, fouling, erosion potential, water quality and tubing material, among others. Typical design cooling water velocities for clean sea water and adjusted for tube material are 2 m/s for aluminum-brass tubes and over 2.4 m/s for titanium. The cooling water temperature increase is frequently between 8 and 14 °C. The cooling water temperature significantly affects the minimum available turbine back-pressure which will be of about 5.1 and 11.9 kPa for inlet water temperatures of 13 and 35 °C, respectively.

The water-cooled surface condenser is the dominant technology used in modern power stations where cooling water is readily available. The flow of large volumes of fresh water and chiefly seawater provided to absorb the heat given off by the condensing steam tends to create biological deposits on the inner surface of the tubes of the heat transfer equipment. Those deposits are commonly known as fouling representing a more complex process in which, besides biological growth, sedimentation, precipitation and sometimes corrosion, are also present giving an important inorganic content to the final layer.

The amount of heat rejected from thermal power stations is significant, representing more than 50% of the total heat input. Apart from 10 to 15% rejected up the stack with flue gas the most of it must be absorbed from condensing the steam in the Rankine cycle.

The heat rejection systems are generally classified as either once-through (open) or closed. In once-through systems, water is withdrawn from a lake, river or ocean and then pumped through the condenser for the abovementioned purposes. The warmer water is then discharged back to its source. In closed loop systems, heat is rejected to the atmosphere usually through the use of cooling towers (wet or dry) or dry systems like air-cooled steam condensers which are gaining in acceptance—in spite of its higher turbine back-pressure (13 - 30 kPa) and lower plant efficiency—due to environmental restrictions.

Once-through cooling systems are in many cases no longer acceptable because of increasingly stringent environmental regulations. Nevertheless, this was the cooling system

type used in both of the power plants in which our research took place, due to the excellent conditions of the particular withdrawal site (Algeciras bay). In Figure 1.1 we can see the diagram of the once-through cooling system at “Los Barrios” power station: Twin shell single pass multi-pressure receiving the steam in downflow.

1.2. Problems Caused by Fouling in Heat Exchangers

Fouling frequently involves a serious problem to achieve optimum operational conditions in heat exchangers-condensers. The economic cost and energy losses associated with this phenomenon are significant and the environmental impact of biocides must satisfy stringent regulations.

Fouling formation is a major concern in industrial heat exchangers. In coastal power stations using seawater for cooling purposes the potential for biological growth is high and fouling is particularly severe in steam condensers, where fouling formed on the inside surface of condenser tubes causes increased heat transfer and frictional resistances reducing heat transmission and causing energy losses. These and other undesirable effects such as promotion of corrosion and flow restriction lead to increased maintenance and adversely affect the efficiency, heat rate, and the economy of the plant.

The formation of fouling in heat exchangers of coastal power plants using sea water for cooling purposes has special economic significance (Brankevich et al., 1990; Bott, 1995 Eguía, 1998). In this type of facilities, fouling is formed inside the condenser tubes, reducing heat transfer between the hot fluid (steam that condenses in the external surface of the tubes) and the cold sink (sea water flowing through the tubes). This fouling has negative consequences in the efficiency of the power plant and therefore in its economic balance (Melo and Bott, 1997).

The problems caused by fouling can be categorized into two different groups: a) difficulties related with the performance of process equipment and b) additional costs.

a) Operational Problems

The formation of deposits on the heat transfer surfaces causes an increasing in frictional pressure drop, due to increased surface roughness and restricted cross-sectional flow area, and, also, an increased heat transfer resistance.

b) Costs due to Heat Exchanger Fouling

Despite the enormous costs associated with heat exchanger fouling, only very limited research has been done to determine accurately the economic penalties due to fouling and to attribute these costs to the various aspects of heat exchanger design and operation.

Capital Expenditure

Excess Heat Transfer Surface Area

Thackery (1980) found that design excess surface area for fouling ranges between 10 – 500%, with an average around 30%. This 30% may correspond to 25% additional capital cost.

Transport and installation costs

As a result of additional surface area, heat exchangers become bigger and heavier. Therefore costs for stronger foundations, provisions for extra space, increased transport and installation costs must be considered.

Capital Costs for Anti-fouling Equipment

These costs include expenses for on-line and off-line cleaning equipments, extra cost for providing no fouling heat exchangers such as scraped surface or fluidised bed heat exchangers, pre-treatment plants, cleaning-in-place equipments, dosing pumps and tanks for anti-fouling chemicals.

Fuel Costs

Costs for extra fuel only occur if fouling leads to extra fuel burning in furnaces or boilers or if more secondary energy such as electricity is needed to overcome the effects of fouling.

Maintenance Costs

Maintenance costs are those for removing fouling deposits and costs of chemicals or other operating costs of antifouling devices. According to Pritchard (1987) and Thackery (1980), about 15% of the maintenance costs of process plants could be attributed to heat exchangers and boilers and 50% of them were probably due to fouling.

Costs due to Production Loss

Because of programmed and unsheduled plant shut-downs due to fouling in heat exchangers, large production losses are produce. These costs are often considered to be the main cost of fouling. For example: Sart and Eimer (1979) state that the loss of production for a one day shut-down of a 1300 MW power plant is about \$US 500000. In addition to production losses during the plant shutdown and start-up, penalties for not keeping to a deadline and the loss customers must be considered.

1.3. Fouling Mitigation Treatments

Many factors influencing fouling need to be addressed at both the design and operational stages. These include operation at the appropriate velocities and temperatures, avoiding dead zones in heat exchangers and ensuring that proper screening of the input water is achieved. Even when all these factors are properly addressed, fouling (and particularly biofouling) may still occur and it may be appropriate to seek mitigation, control and cleaning methodologies to maintain this fouling at acceptable levels.

The emphasis will be on mitigation and control, which are essentially "on-line" processes, carried out either continuously or intermittently. The objective of the mitigation is the reduction or removing of fouling located on the heat transfer surfaces or at least to maintain this fouling at acceptable levels from an efficiency point of view.

A wide variety of methods have been used for fouling treatment and control and these are summarised in Table 1.1. These treatments could be chemical, physical, biological, etc. The methods applied in this study are shown in Table 1.1.

Within the chemical mitigation, the emphasis will be on the biocides, because these are the most common approach to biofouling problem in industrial water system. Biocides have been subject to numerous reviews and will not be discussed exhaustively in this chapter. However, the methods applied in this work shall be briefly discussed.

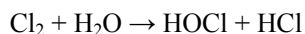
Table 1.1. Methods of biofouling mitigation and control

Mitigation	Chemical
	<i>Oxidising biocides</i> • chlorine* • ozone • bromine • peracetic acid* <i>Non-Oxidising biocides</i> • acroleine • glutaraldehyde • isothiazolones • quaternary ammonia compounds* • biodispersants
	Physical
	• flushing • backwashing • abrasive sponge balls • brushing • hot water, steam • irradiation • ultrasonic energy • ultra-violet light*

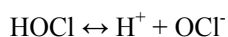
Chlorine

Chlorine has been a preferred biocide for many years on account of its effectiveness and its relatively low cost. It has been used particularly in once through cooling water systems where large volumes of water are used, in order to minimise the cost of treatment. It is effective against micro- and macro-organisms.

Free chlorine reacts with water to form two acids. The reactions are:



The hypochlorous acid ionises in water according to the equilibrium:



The presence of hypochlorous acid as opposed to the hypochlorite ion, is responsible for the biocidal activity. Hypochlorous acid is estimated to be twenty times more effective as a micro-biocide than the hypochlorite ion.

Chlorine could be applied in different ways: liquid (compressed gas), sodium hypochlorite solutions, calcium hypochlorite powder, organic compounds – chlorine donors and *in situ* generation, e.g. electrolytically generated from sea water.

Because the chlorine can react with many organic substances found in water systems, there is a “chlorine demand” that will affect the biocidal activity of a particular dose. It is usually necessary to add chlorine in excess in order to get a free residual of chlorine in the system to ensure efficient biocidal action. Some comments on chlorine treatment programmes are contained in Table 1.2 (Betz Laboratories Inc., 1976).

Table 1.2. Effectiveness of chlorine addition

Programme	Remarks
Continuous chlorination – free residual	Most effective Most costly Not always technically or economically feasible due to high chlorine demand
Continuous chlorination – combined residual	Less effective Less costly Inadequate for severe problems
Intermittent chlorination – free residual	Often effective Less costly than continuous chlorination
Intermittent chlorination – combined residual	Less effective Less costly

The final choice of the chlorine program depends on the type of biofouling which can be expected.

The concentration levels are usually expressed in parts per million mass. The general intention is to have a residual concentration of 0.1 - 0.2 mg/l beyond the last point requiring protection.

The selection of chlorine as a control medium causes no problems when titanium is used as a material of construction but could cause considerable trouble with many other materials. For copper or Superduplex alloys, a level of 0.5 mg/l of chlorine may be acceptable but care should be taken to avoid excursions in concentration to mg/l levels.

Film-Forming/Dispersing Additives: Mexel®432

Mexel®432 is an organic antifouling product, which has recently appeared on the market and aim to treat seawater systems against inert deposits, fouling by living organisms, sessile bacteria and corrosion.

This product has the following characteristics. Its chemical name is alkylamino -3 aminopropane and its formulation is 1.7% (Alkyl amino)-3 aminopropane active ingredient (alkyl as in fatty acids of coconut oil). This product has been registered by EPA as aquatic pesticide (SLN 950007, EPA 69100).

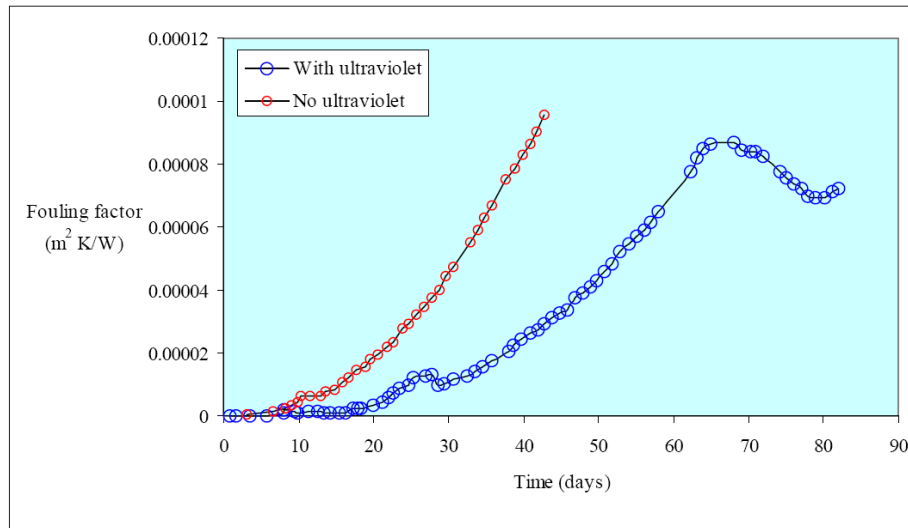
This hydrocarbon compound is a mixture of aliphatic hydrocarbons, with alcohol and amine functionality, that work in the presence of dissolved oxygen in waters a neutral pH (i.e. over 6). It is an anti-fouling material that acts as a corrosion inhibitor and scale dispersant as well as having activity against freshwater and saltwater mussels and barnacles (Giamberini et al., 1994; Krueser et al., 1997). The amines act as surfactants, or “filming amines,” and adhere to wetted metal, plastic, concrete, and glass surfaces to form a film, which is hydrophobic, through which biofouling organisms cannot form an attachment. This preventive activity avoids mussel infestation by repelling veliger settlement and adhesion to clean surfaces. This product is used for control of molluscs, including zebra mussels and clams, in industrial water systems, by retarding byssal thread formation, adhering to and damaging gill surface membranes, and dispersing mussels. Thus, it prevents new infestations and gradually disperses existing infestations, and is used primarily to prevent infestations in a previously cleaned system.

Initial application early in the season prior to veliger settlement is most effective, with continuation of daily dosing throughout warm weather. Treatment shows effects within a few days, but continuing treatment is required. Monitor treatment efficacy with bioboxes located at critical points in the system.

Mexel®432 is recommended by the manufacturer at a dose of 50 l per 1000 m² of area every month in two to three hours injection a week. Dosage is calculates according to the area to be treated rather than the water flow rate. However, 2 mg/l would be requires to kill off mussels completely (Mexel S.A.).

Ultraviolet Light (UV)

UV light is a physical, non-intrusive method of disinfecting water, which is very effective in controlling micro-biological attack by sulphur and ion-reducing bacteria, which results in corrosion underneath the biofilm (Clark, 1984). However, it would not be an effective way of controlling biofouling caused by such organisms as mussels and barnacles because they are protected by their shells and the UV treatment don't have either a consistent impact on the biofilm community (Pozos et al., 2004). Tests have indicated (Seki et al., 1985, 1986) that ultra-violet radiation can subdue marine biological growth in the early stages by as much as 25%. However, around 0.5% of marine micro-organisms appear to be UV resistant and can grow on surfaces to give high fouling as illustrated in Figure 1.2. Thus, the use of UV may not be very effective unless used with some other technique. Furthermore, the surface to be irradiated must be “visible” to the light source.



Seki et al., 1985, 1986.

Figure 1.2. Comparison of fouling resistance development with and without ultraviolet radiation.

UV light is in the region of the electromagnetic spectrum between visible light, which has wavelengths greater than 400 nm, and X-rays (less than 100 nm). UV has three spectral bands: (1) 100 to 280 nm, which is short-wave UV; (2) 280 to 315 nm, middle-wave UV; and (3) 315 to 400 nm, long-wave UV. Disinfection by UV light is performed by short-wave UV, which is absorbed by the reproductive mechanisms of bacteria and viruses, the genetic material (DNA/RNA) is rearranged and they can no longer reproduce. They are therefore considered dead and the risk of disease has been eliminated.

Ultraviolet light is most effective when the water has already been partially treated, and only the cleanest water passes through the UV flow chamber. Regular low-pressure UV lamp systems should use 40 – to 60 UV lamps per million gallons per day (3790 m³/d). The power requirement for each lamp ranges from 65 to 80 W. The intensity of the UV light emitted by a lamp is expressed in mW/cm². UV dose is calculated by multiplying the intensity by the time of exposure and expressed as mWs/cm². A dose ranging from 25 to 50 mW s/cm² will be effective in water with low turbidity

1.4. Environmental Impacts of the Treatments

The performance of the cooling water systems create several detrimental effects on the natural environment: heat discharges to the atmosphere, heating of receiving aquatic-environments, suction of organisms into water intakes, alteration of the receiving environment by chemical discharges. Other possible harmful effects resulting from the choice of some cooling systems are related with the aesthetics and noise (wet cooling towers). For dry cooling towers and condensers, in addition to the two previously mentioned aspects, there is also the possible dissemination into the air of corrosion products from the heat exchange surface area.

Between these impacts, the emphasis will be on the alteration of the receiving environment by chemical discharges because it is the topic of this section.

The water withdrawn for cooling purposes may sometimes be the cause of chemical releases into the receiving environment. The following may be concerned in particular:

- reagents used to avoid the scaling of cooling systems equipped with cooling towers;
- reagents used to fight against biological developments, reaction products of some of them;
- iron sulphate anti-corrosion treatments to protect, in some cases, copper alloy condensers;
- corrosion products of heat exchangers and piping.

Environmental Impact of Chlorine

When chlorine reacts with some organic matter, it may lead to the formation of organohalogenated substances (mainly bromoform in seawater). These chemicals products of chlorination represent toxic and carcinogenetic pollutants and are persistent in the environment. Furthermore, these products can enter in the food chain.

As results of these concerns the use of chlorine is being restricted either on a voluntary basis or through legislation.

Because of the concern for the environment it may be necessary to remove the chlorine (dechlorination) from the water before discharge. Dechlorination methods may be listed as follows (EPRI, 1984):

1. The use of reducing chemicals e.g. sulphur dioxide, sodium sulphite as sodium thiosulphate. The technique is likely to involve elaborate dosing equipment, e.g. automatic control, in order to maximise effectiveness at minimum cost.
2. Adsorption onto granular activated carbon (i.e. similar to that used for water treatment).

Environmental Impact of Mexel®432

Mexel®432 is considered by its manufacturers to have no impact on the environment, and, is no toxic to mammals. In addition, It is also biodegradable, not oxidant and non toxic, however some studies have evaluated the effect of this compound on aquatic organism and they have proved the opposite thing. For example, *Dreissena polymorpha* have shown that continuous exposure of 3.5 mg/l residual Mexel®432 produced 100% mortality after 20 h (Khalanski, 1993) and at intermittent injection of 6 mg/l during 3 h per day induced 100% mortality after 15 d of exposure (Giamberini et al., 1994).

Little information is available on the biodegradation of this chemical in marine environment.

Environmental Impact of UV

UV light is a physical, non-intrusive method of disinfecting water that doesn't affect the chemistry, pH, smell or color.

Therefore, Ultraviolet light is a natural, cost effective, environmentally friendly disinfection process which has, however, the disadvantage of being a complementary technology in the mitigation of the biofouling.

2. TECHNIQUES FOR MONITORING AND CHARACTERIZATION OF FOULING

2.1. In-situ Studies: Pilot Plant for Monitoring the Fouling

Taking into account that the type and rate of fouling will be dependent on the specific characteristics of the cooling water, climatology and other operating conditions of every ship or industrial plant *in situ* experimental studies of fouling process have been carried out using a portable pilot plant (Figure 2.1) specially designed to simulate fouling development in heated tubes, as might occur in a condenser or heat exchanger, as close as possible to the real systems. In addition, the pilot plant –as a side-stream monitoring device– will provide the necessary information to predict the fouling layer formation under controlled conditions and different biocide dosage patterns for, finally, optimizing its application schedule (Casanueva et al., 2003).

To facilitate transportation the whole plant has been installed inside a 20 ft standard container.



Figure 2.1. Pilot plant located at two different power stations.

All the equipment is fixed to the soil and/or walls by means of suitable bedframes and supports, in order to avoid damage during transportation. The shell-and-tube heat exchanger is faced to the container door so that the tubes can be easily removed for replacement or maintenance operations. The container base has been reinforced to support the total weight of the equipment. It also has a light slope towards one of the corners, where there is a drainpipe for evacuation of any liquid spill. The floor is protected with a neoprene carpet. The container is also equipped with a compact air conditioner.

A diagram of the experimental system is presented in Figure 2.2, where the different circuits can easily be followed.

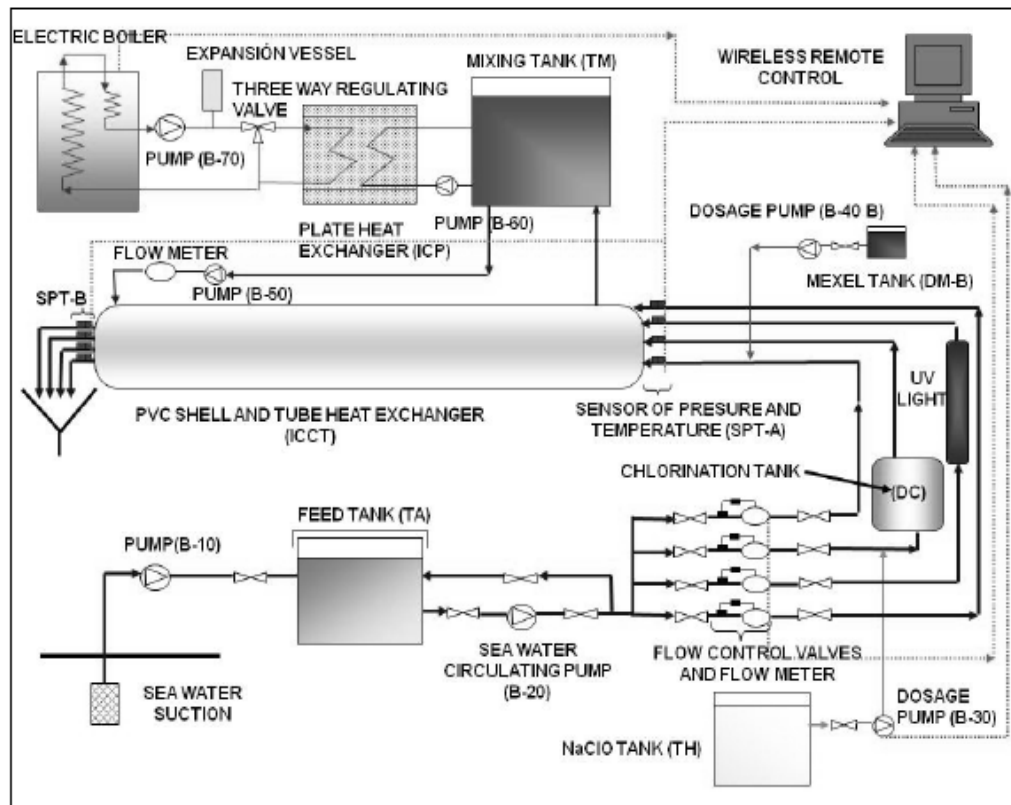


Figure 2.2. Diagram of pilot plant experimental system.

An exhaustive list of the different components of the pilot plant is the following:

- Seawater feed system which consists of feed pump (B-10) and seawater feed tank (TA).
- Seawater circulation system to force sea water through all the circuit, and consisting of seawater circulation pump (B-20), chlorination tank (DC), UV reactor and the distribution system for supplying the required flow to each test tube.
- PVC shell-and-tube heat exchanger (ICCT) 3100 mm in length and specially designed to avoid galvanic corrosion.
- Biocide dosage system, including the biocide tanks (TH) and (DM) and the dosage PLC controlled pumps (B-30) and (B-40).
- External heating system for the PVC heat exchanger, including the mixing tank (TM), the plate heat exchanger (ICP), the electric boiler (CE), the circulating pump for the primary heating fluid (B70), the circulating pump for the secondary fluid (B60) and the expansion pressurized vessel and circulating pump for the tertiary fluid (B50).
- Control and data acquisition systems: for the control, processing and measurement of flow, temperature, chlorine (REDOX potential) and pressure drop.
- Cabinet for electrical and power components, PLCs, displays.
- Cabinet for computer, monitor, keyboard and printer.

Advantages of this pilot plant for *in-situ* field experimentation include the fact that the adaptability and full automation of the pilot plant make possible a wide range of experimentation possibilities.

The design of the main heat exchanger and the whole plant makes it easy to carry out studies with different tube materials, diameters, biocides and dosage patterns, which is very useful for optimizing biofouling control procedures. Wireless remote control, monitoring and data transmission of the pilot plant can be carried out via a modem.

Different methods for monitoring the fouling progression have been reported by diverse authors. Throughout our research effort we have made use of those which are next explained.

2.2. Indirect Methods for the Quantification of Fouling

In our study, indirect measurement of transport properties -heat transfer resistance and flow resistance- are employed and changes in them are computed and monitored. Our pilot plant uses a SCADA system to calculate the friction factor and the resistance to the heat transmission, according to the current values of the on-line measured variables (pressure drop, flow rate and temperature), and represents graphically their evolution *versus* time for every tested tube.

The overall heat transfer resistance (sum of conductive and convective resistances) is expressed by the reciprocal of the overall heat transfer coefficient and will be determined for each tube in accordance with the following equation:

$$R = U^{-1} = \frac{A_o \cdot [(t_{hwo} - t_{cwi}) + (t_{cwo} - t_{cwi})]}{\frac{\pi}{4} \cdot d_i^2 \cdot v \cdot \rho \cdot C_p \cdot (t_{cwo} - t_{cwi}) \cdot \ln \frac{t_{hwo} - t_{cwi}}{t_{hwi} - t_{cwo}}} \quad (\text{m}^2 \cdot \text{K} / \text{W})$$

By calculating the difference in R between fouled and clean conditions the fouling resistance R_f can be determined as fouling progresses.

Increased frictional resistance is, also, measured as the effective diameter of the tube diminishes and roughness increases. This will result in an augmented pressure drop of water through the tubes. Frictional resistance is conveniently expressed by the Darcy dimensionless friction factor.

$$f = \frac{2 \cdot d_i \cdot \Delta P}{L \cdot \rho \cdot v^2}$$

Nomenclature

A_o	outside surface of tube (m^2)
C_p	specific heat capacity of seawater at bulk temperature ($J\ kg^{-1}\ K^{-1}$)
d_i	inside diameter of tube (m)
f	friction factor (dimensionless)
L	tube length (m)
R	overall heat transfer resistance referred to outside surface of tube wall ($m^2\ K\ W^{-1}$)
R_f	fouling heat transfer resistance ($m^2\ K\ W^{-1}$)
t_{cwi}	cooling water inlet temperature (K)
t_{hwi}	heating water inlet temperature (K)
t_{cwo}	cooling water outlet temperature (K)
t_{hwo}	heating water outlet temperature (K)
U	overall heat transfer coefficient referred to outside surface ($W\ m^{-2}\ K^{-1}$)
v	cooling seawater velocity ($m\ s^{-1}$)
Δ_p	pressure drop along tube length ($N\ m^{-2}$)
ρ	seawater density ($kg\ m^{-3}$)

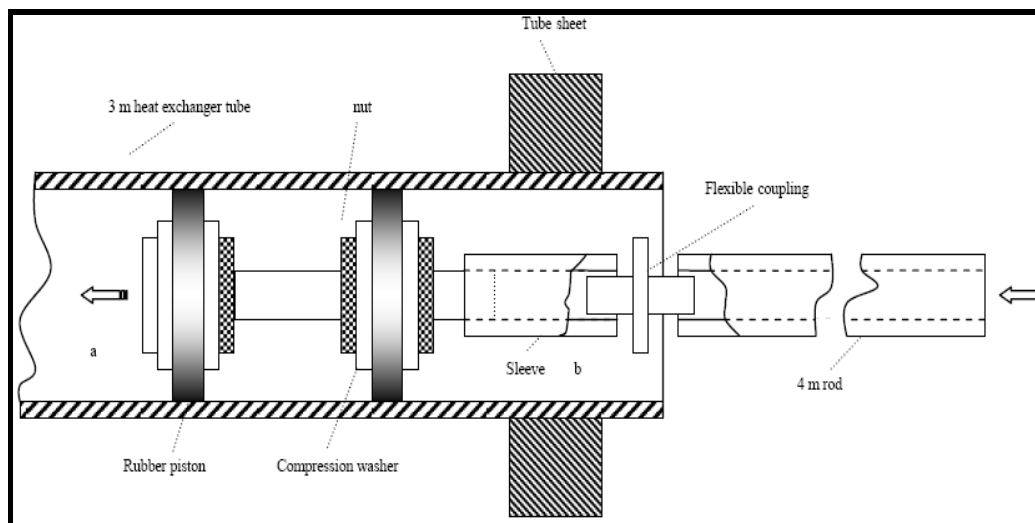
2.3. Direct Methods for the Characterization of Fouling

These concern the determination of the quantity or properties of the biofilm directly from the fouled support. Frequently, these type of procedures can be catalogued either as *non-destructive methods*, like direct observation using optical microscope or determining some specific activities, or as *destructive methods* involving previous biofilm removal or direct measurement over the support (Lazarova and Manen, 1995; García-Morales, 1997).

A stainless steel tool, which is shown schematically in Figure 2.3, has been designed to collect the inner deposit, allowing us to carry out direct measurements of the fouling layer. Even though this is done at the end of the experiment, it provides valuable information and gives us the possibility of contrasting with the indirect measurements and also to check the sensitivity of the indirect method used. We have employed this device as follows and good results have been achieved.

Prior to opening the tubes that were tested, heating is turned off and the cooling water flow rate is gradually lowered. In this way we prevent the fouling from drying and from detachment. Previous tests had confirmed, by filtering the discharged water, that any significant mass of film was eliminated, following the aforementioned procedure.

Once we have removed the set of sensors connected at the end of each tube, we can tighten the nuts, and compress the rubber rings until they fit properly into the inner surface of the tube. The tube is then completely cleaned (zone b) by the double rubber piston as it is pushed through the 3.2 m heat exchanger tube, by means of the 4 m rod. Owing to the second piston, any remaining fouling layer after the first one has functioned is collected and also acts as a guide maintaining the concentric position of the set, not disturbed by the pushing rod due to the flexible coupling. The deposit is easily collected (zone a) at the other end of the tube for the subsequent laboratory measurements.



Casanueva et al., 2003.

Figure 2.3. Tool for fouling removal.

2.3.1. Physico-Chemical Methods

In this work we characterize the heat exchanger biofouling by physico-chemical methods and the first time we introduce a point of view biological through the study of bacterial species responsible of biological fraction of deposit fouled.

By means of a double rubber piston that fitted perfectly into the inside of the tubes, they biofouling samples were completely cleaned along their 3.2 m length. The subsequent laboratory measurements, including:

- Fouling wet volume, measure after 4 h settling of the collected fouling in an Imhoff cone
- Fouling mass of total volatile and inert solids (gravimetric analysis).
- Layer thickness, calculated as the ratio between the wet volume and the inner surface of the tube. Although this is a simple method of measurement, a good correlation between film mass and thickness was obtained.

For characterizing the metallic composition different techniques were used as: Elemental analysis (CHNS). The instrument and the processes used are:

CHNS

The CHNS instrument can determine the carbon, hydrogen, nitrogen, sulphur and oxygen for homogeneous organic material.

High temperature combustion is used as the means of removing the elements form the material. In the combustion process, a nominal 2 mg sample is encapsuled in a tin capsule. The sample is placed in the sample loading chamber and held there until a dose of oxygen has been released. The sample is then dropped into the furnace at the same time the oxygen arrives. The sample is combusted by the heated oxygen-rich environment.

The products of combustion in a CHNS analysis are CO₂, H₂O, N₂ and SO_x. The gases, which are carried through the system by helium carrier, are swept through the oxidation tube packed with either tungsten trioxide (which adds oxygen) or copper sticks (which removes oxygen), to complete the conversion to SO₂. The H₂O is swept through the non-dispersive infrared absorption detection system where it is measured, Anhydrone then removes the H₂O and the remaining gases are swept through the SO₂ and CO₂. Lecosorb and Anhydrone leaving the helium carrier gas and N₂ remove the CO₂ and the H₂O. The N₂ is measured by thermal conductivity. Adjustments for blank, calibrations, and weights are applied to the final integrated signal and the answers are displayed as weight percent carbon, hydrogen, nitrogen and sulphur.

Table 2.1 Shows the precision of the instrument

Element	Sigma	Coefficient of variation
Carbon	0.001	0.3
Hydrogen	0.01	1.0
Nitrogen	0.02	0.3
Sulphur	0.02	1.0

2.3.2. Molecular Methods

The introduction of molecular biology to microbial ecology has had a strong impact. The subject of prokaryotic biodiversity in natural samples such as biofilm communities has received new and substantial attention with the development of molecular techniques to describe and identify the microbial components of natural communities. PCR amplification and the cloning of diagnostic molecules, mostly the 16S rRNA genes, permits extensive studies of the microbial diversity of ecosystems without the bias imposed by pure-culture techniques.

In the present study, the bacterial community associated with a biofilm from a marine water of a heat exchanger tubes in a pilot plant was characterized by analysis of 16S rDNA gene libraries. Molecular identification and phylogenetic reconstructions based on 16S rDNA gene sequences were performed allowing compare the bacterial community with different biocides treatment used in coolant water system. The bacterial diversity of the biofilm samples was studied with the protocol (Schabereiter-Gurtner et al., 2001) showed in Figure 2.4

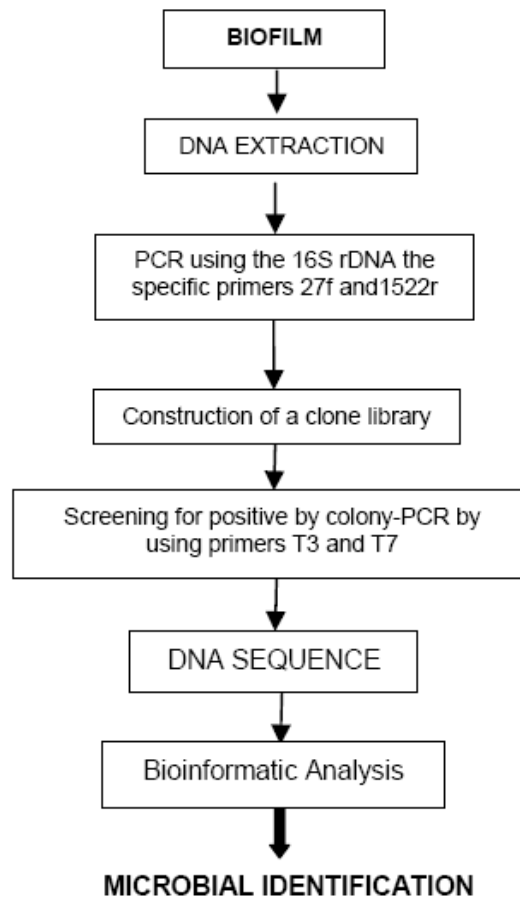


Figure 2.4. Schematic diagram of the experimental steps to analyze the structure of microbial communities and the phylogenetic affiliation of biofilm community in heat exchanger tube.

Biofilm Sample Harvest and Pretreatment

This step, crucial for all subsequent analyses, is often the main source of problems. Sampling site, time and condition used during transport to the laboratory, storage and later processed are important parameters to keep in mind to obtain representative results of the composition of the bacterial community. Also, the marine samples present an added problem: the presence of salts and inhibitors of the polymerase make difficult to obtain amplifications of DNA fragments of the appropriate size and enough quality. Biofilm samples, including bacterial cells deposited after each treatment, were collected using a suitable device (Figure 2.3) in a sterilized vessel and quickly stored at -80 °C until sample processing. Samples were thawed at 4 °C, centrifuged at 6000 x g and the pellet was washed, for three times, in a volume of MilliQ sterilized water, before DNA extraction. All biofilm samples were harvested between months of July to September.

Genomic DNA Extraction and PCR Amplification of Bacterial 16S rDNA Gene

Genomic DNA from samples was extracted using UltraClean Microbial DNA Isolation Kit (MOBIO Laboratories, Inc. Carlsbad, CA). DNA integrity was analyzed in a 0.8% agarose gel. Genomics DNA was diluted in water to a final concentration of 40 - 50 ng/μl for PCR amplification. Quality of purified genomic DNA was estimated spectrophotometrically by measuring 260/280 absorbance relationship (Figure 2.5) Samples that showed a relationships under 1.7 were discarded.

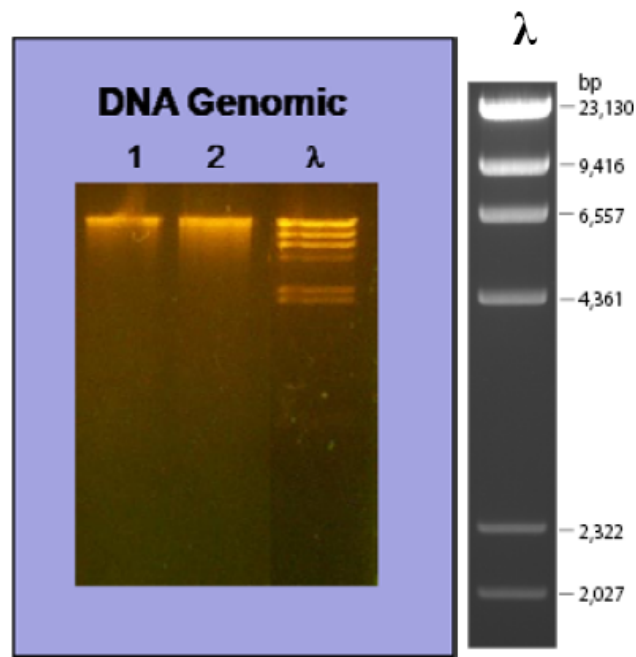


Figure 2.5. Recovered genomic DNA electrophoresis. 1, 2: DNA extracted from biofilms community in heat exchanger tube; λ: Lambda Hind III molecular weight standard. The numbers indicate the size in bases pairs of each one of the standards.

Polymerase Reaction Chain (PCR) is used to amplify specific regions of a DNA strand (the DNA target). This method permits PCR targeting smaller primer binding regions, and is particularly useful to amplify unknown, but conserved, DNA sequences, such as the 16S rRNA gene. Actually, this gene is most useful for detection, identification and classification of bacteria. 16S rDNA gene was amplified using 27f and 1522r universal primers (Table 2.2), which are specific for universal conserved bacterial 16S rDNA sequences. All amplifications were performed in a MiniOpticom PCR System (BioRad, Munich, Germany). PCR products were analyzed by electrophoresis in a 0.8% agarose gel in TAE x 0.5 buffer to check the presence of a fragment of about 1500 bp (Figure 2.6).

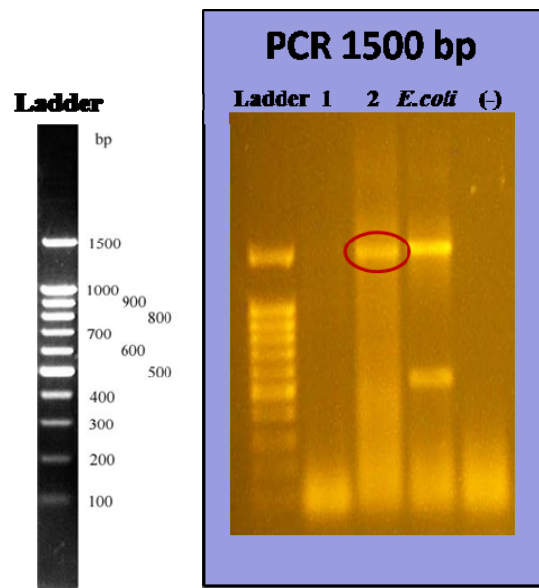


Figure 2.6. Agarose gel showing the PCR product obtained after amplification of isolated DNA from biofilms. Lane 1: No amplified sample, lane 2: sample with 1.5 kbp band amplified, *E.coli*: positive control and (-): negative control. Ladder: 1,5kbp +100bp molecular weight standard.

Table 2.2. Oligonucleotides used in this study

Primer	Sequence 5' 3'	Position ¹	Target	Reference
27f	AGA GTT TGA TCC TGG CCA G	27	Bacteria	Weisburg et al., 1991
1522r	AAG GAG GTG ATC CAG CCG CA	1522		González et al., 2003
341f	TAC GGG AGG CAG CAG	341		Muyzer et al., 1993
T7	TAA TAC GAC TCA CTA TAG GG	-	Vector pCR 4	Invitrogen (Carlsbad, CA, USA)
T3	ATT AAC CCT CAC TAA AGG GA	-		

¹Relative position in 16S rRNA in *Escherichia coli*.

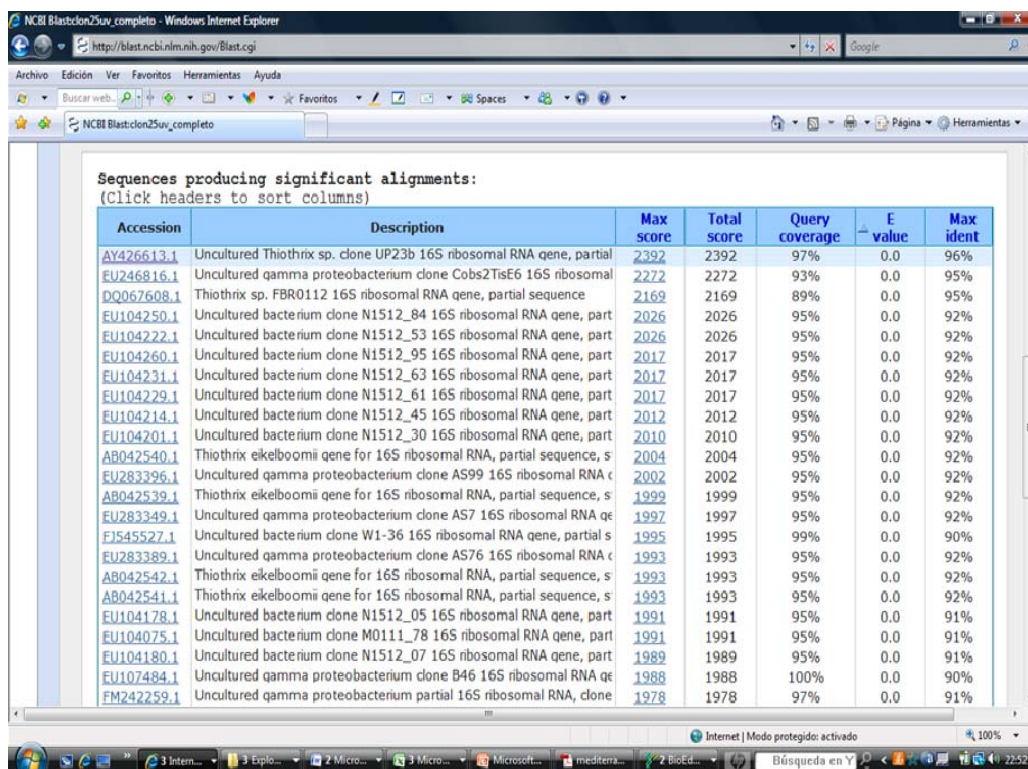
Construction of Clones Libraries and Sequencing of 16S rDNA

Proper PCR product (1500 bp) was sliced from the gel, purified using a GENECLAN Turbo PCR kit (Qbiogen, Carlsbarg, CA, USA) and cloned into pCR®2.1-Topo® (Invitrogen), following the instruction manual. The ligation products were transformed into chemically competent *Escherichia coli* Top-10 cells (Invitrogen Corporation). Transformed cells were screened for the presence of recombinant plasmids and positives plasmids were purified for a subsequent sequenciation step of the insert. The nucleotide sequences of inserts were determined by using the ABI PRISM dye terminator cycle-sequencing ready-reaction kit

(Perkin-Elmer), according to the manufacturers' instructions using T3 and T7 universal primers (Table 2.2), in an ABI Prism 3100 Genetic Analyser (Applied Biosystem). 255 full sequences of 16S rDNA genes of about 1500 bp were obtained and later *in silico* analyzed

Bioinformatic Analysis

Closest matching sequences in the Genbank database were identified using BLAST (Basic Local Alignment Search Tool) (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) from NCBI (National Center for Biotechnology Information) (Figure 2.7)



Sequences producing significant alignments:
(Click headers to sort columns)

Accession	Description	Max score	Total score	Query coverage	E value	Max ident
AY426613.1	Uncultured Thiothrix sp. clone UP23b 16S ribosomal RNA gene, partial	2392	2392	97%	0.0	96%
EU246816.1	Uncultured gamma proteobacterium clone Cobs2TisE6 16S ribosomal	2272	2272	93%	0.0	95%
DQ067608.1	Thiothrix sp. FBR0112 16S ribosomal RNA gene, partial sequence	2169	2169	89%	0.0	95%
EU104250.1	Uncultured bacterium clone N1512_84 16S ribosomal RNA gene, part	2026	2026	95%	0.0	92%
EU104222.1	Uncultured bacterium clone N1512_53 16S ribosomal RNA gene, part	2026	2026	95%	0.0	92%
EU104260.1	Uncultured bacterium clone N1512_95 16S ribosomal RNA gene, part	2017	2017	95%	0.0	92%
EU104231.1	Uncultured bacterium clone N1512_63 16S ribosomal RNA gene, part	2017	2017	95%	0.0	92%
EU104229.1	Uncultured bacterium clone N1512_61 16S ribosomal RNA gene, part	2017	2017	95%	0.0	92%
EU104214.1	Uncultured bacterium clone N1512_45 16S ribosomal RNA gene, part	2012	2012	95%	0.0	92%
EU104201.1	Uncultured bacterium clone N1512_30 16S ribosomal RNA gene, part	2010	2010	95%	0.0	92%
AB042540.1	Thiothrix eikelboomii gene for 16S ribosomal RNA, partial sequence, s	2004	2004	95%	0.0	92%
EU283396.1	Uncultured gamma proteobacterium clone AS99 16S ribosomal RNA c	2002	2002	95%	0.0	92%
AB042539.1	Thiothrix eikelboomii gene for 16S ribosomal RNA, partial sequence, s	1999	1999	95%	0.0	92%
EU283349.1	Uncultured gamma proteobacterium clone AS7 16S ribosomal RNA ge	1997	1997	95%	0.0	92%
FJ545527.1	Uncultured bacterium clone W1-36 16S ribosomal RNA gene, partial s	1995	1995	99%	0.0	90%
EU283389.1	Uncultured gamma proteobacterium clone AS76 16S ribosomal RNA c	1993	1993	95%	0.0	92%
AB042542.1	Thiothrix eikelboomii gene for 16S ribosomal RNA, partial sequence, s	1993	1993	95%	0.0	92%
AB042541.1	Thiothrix eikelboomii gene for 16S ribosomal RNA, partial sequence, s	1993	1993	95%	0.0	92%
EU104178.1	Uncultured bacterium clone N1512_05 16S ribosomal RNA gene, part	1991	1991	95%	0.0	91%
EU104075.1	Uncultured bacterium clone M0111_78 16S ribosomal RNA gene, pari	1991	1991	95%	0.0	91%
EU104180.1	Uncultured bacterium clone N1512_07 16S ribosomal RNA gene, part	1989	1989	95%	0.0	91%
EU107484.1	Uncultured gamma proteobacterium clone B46 16S ribosomal RNA ge	1988	1988	100%	0.0	90%
FM242259.1	Uncultured gamma proteobacterium partial 16S ribosomal RNA, clone	1978	1978	97%	0.0	91%

Figure 2.7. Example of the hit list from a BLASTN report, which presents blast scores. The BLASTN program allows users to input either a nucleotide sequence and to search in a nucleotide sequence database. The program returns a list of the sequence “hits”, alignments to the query sequence, and statistical values.

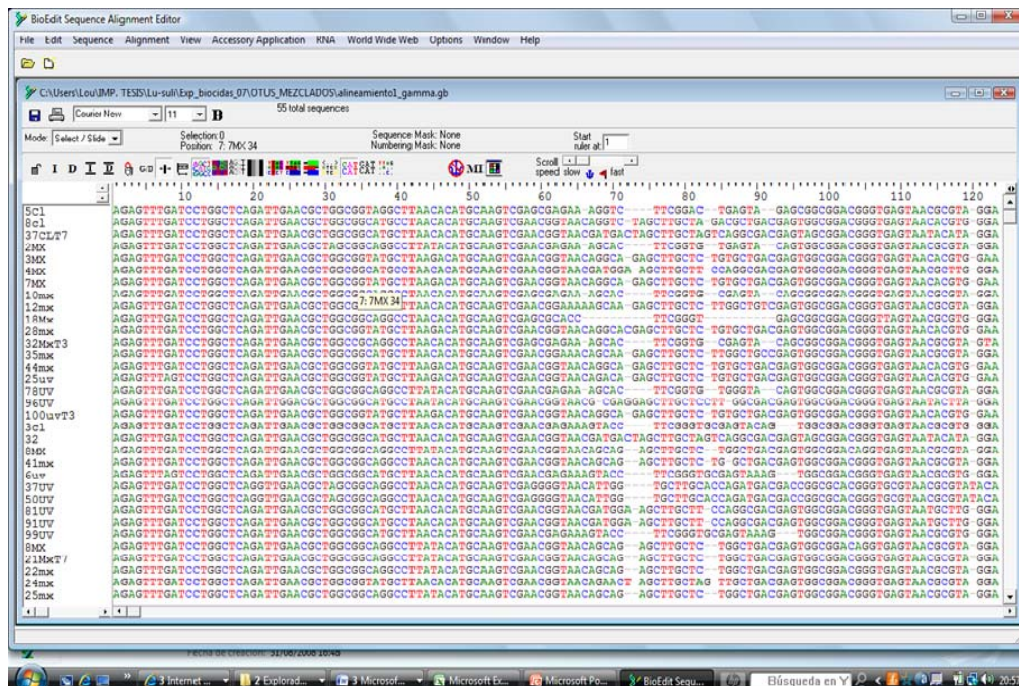


Figure 2.8. Example of DNA multiple sequence alignment performed with Bioedit software (v. 7.0.5.3) (Hall, 1999) showing, in same colour, identical nucleotide in each of the compared sequence. The lack of a nucleotide is represented with a gap. This is a crucial step to obtain a Sequence Identity Matrix.

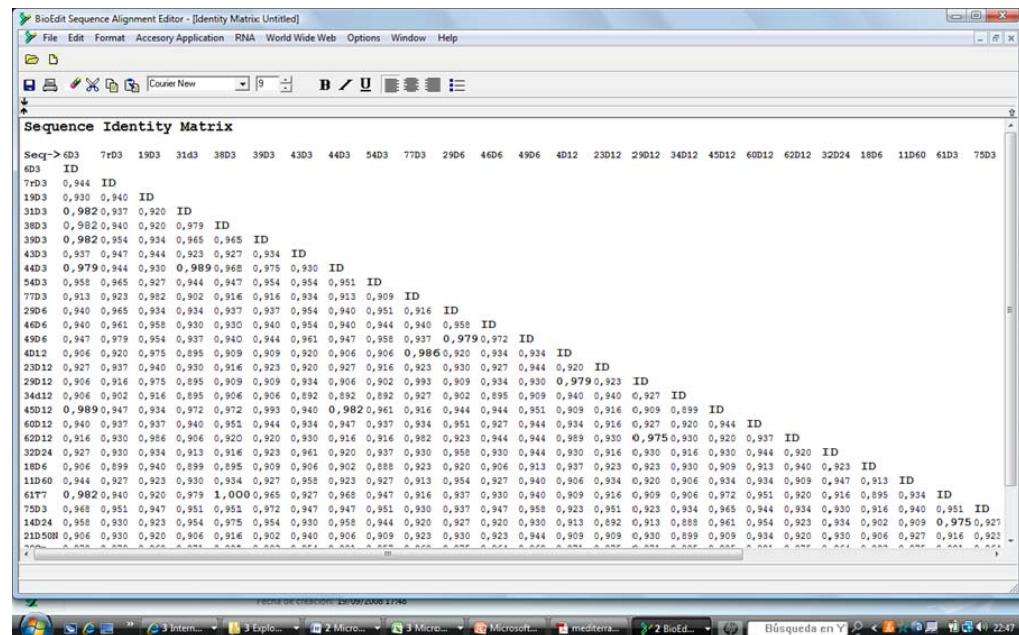


Figure 2.9. Example of sequence identity matrix reconstructed from nucleotide sequences indicating the similarity among clones obtained from biofilms community in heat exchanger tube.

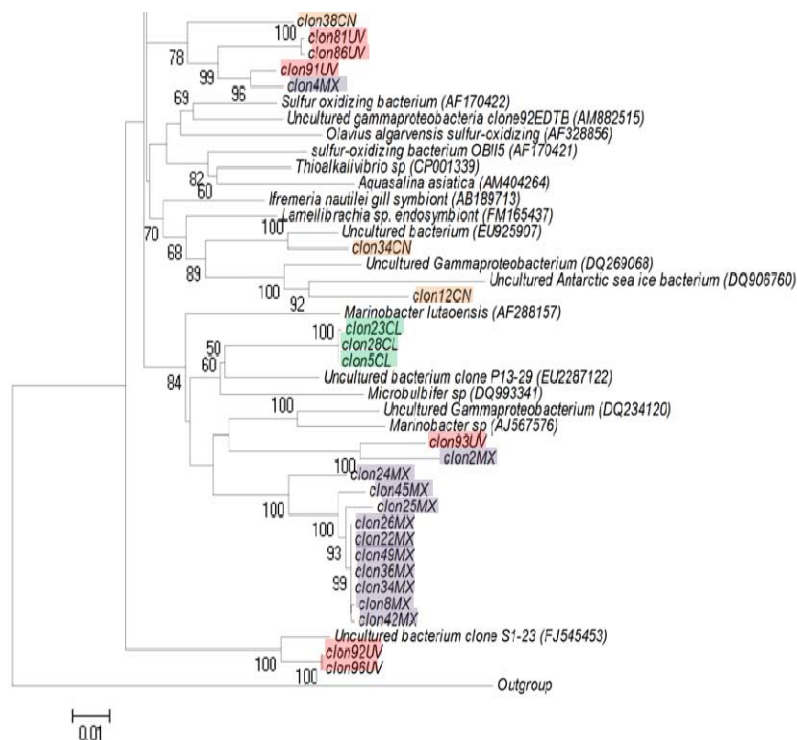


Figure 2.10. Phylogenetic analysis of heat exchanger biofilm showing affiliation with clones grouping within the γ -Proteobacteria. Filled red squared indicate clones retrieved with UV treatment, lilac squared with Mexel @432 treatment, green squared with chlorine treatment and brown squared indicate clones retrieved untreated (control sampled).

Sequences homology was analyzed using ClustalW (Figure 2.8) (Thompson et al., 1994) and Bioedit software (v 7.0.8) (Hall, 1999). A Sequence Identity Matrix (Figure 2.9), where the homology percentages among the compared sequences appear tabulated, was generated. Sequences with homology greater than 98% were grouped together and considered to belong to the same OUT (operational taxonomic unit). Phylogenetic trees were performed with the software MEGA4 (Tamura et al., 2007), using Neighbor-Joining method and Kimura 2-parameters model of distance analysis, and 1 000 bootstrap replications were assessed to support internal branches. Bootstrap values above 50% were included on the trees (Figure 2.10).

The sequence phylotype diversity and sampling efficiency of the libraries were analyzed by the rarefaction curve method using the EstimateS software (<http://viceroy.eeb.uconn.edu/EstimateS>). When rarefaction curve reaches a plateau, the library could be considered 'large enough' to provide an unbiased estimate of OTU richness (Kemp and Aller, 2004).

2.4. Modelization for Fouling Progression

The modelization of fouling deposition has great relevance due the negative effect that the fouling layer over the condenser performance (Epstein, 1981; Bott, 1995). A good model

could be useful for designer and for condenser operators. Therefore, several models have been proposed in literature. Zubair et al. (2000) present a review of models valid to predict some fouling processes occurred in various industrial practices: linear, power-law, falling-rate and asymptotic models. For modelling the fouling that grows in condenser most of authors propose asymptotic models, since when the steady state is reached an asymptotic value of thermal resistance is achieved. One of the first and simplest models to explain the fouling process was put forward by Kern and Seaton (1959) that proposed the following expression:

$$R_f = R_{f\infty}(1 - e^{-kt}) \quad (1)$$

Later, in 1973, Konak proposed a generalised equation for asymptotic fouling based on the driving force concept for deposit development. They suggested that the driving force were the difference between the asymptotic fouling resistance and the fouling resistance at time t .

$$\frac{dR_f}{dt} = k(R_{f\infty} - R_f)^n \quad (2)$$

when $n = 1$, then integration of Konak (1973) equation is equivalent with Kern and Seaton model. Both models consider that the rate of fouling growth is always decreasing from the clean tube until the asymptotic fouling is reached. Nevertheless, experimental evidence shows that fouling development process follows a sigmoidal curve, with a initial period of low rate and after of a induced time, suffer a exponential growths up to achieve the stabilization phase.

The authors have recently published a research work that deals with the modelization for the process of fouling deposition (Nebot et al., 2007). The proposed model permits to predict the maximum value of the thermal resistance of the fouling layer in different conditions. As application and validation of the model performances, the effect of water velocity and material tube has been essayed and modelised. The experimental work was accomplished in a pilot plant located in “Los Barrios” coal power plant, (Bay of Algeciras, Southern tip of Spain).

A good model must have the next characteristics:

- Fit to experimental data,
- To be simple as in its mathematical equation than in its application to experimental data,
- To have physical significance, and so it will be possible to interpret the values of the kinetic parameters of the model.

In the Figure 2.11 is observed the typical sigmoidal shape of the evolution curve of thermal resistance during the fouling deposition. In the induction period, the curve suffers weak fouling accumulation, later it can be seen a quickly growth of the fouling layer, and finally a stabilization period is reached when growth and detachment rates are similar.

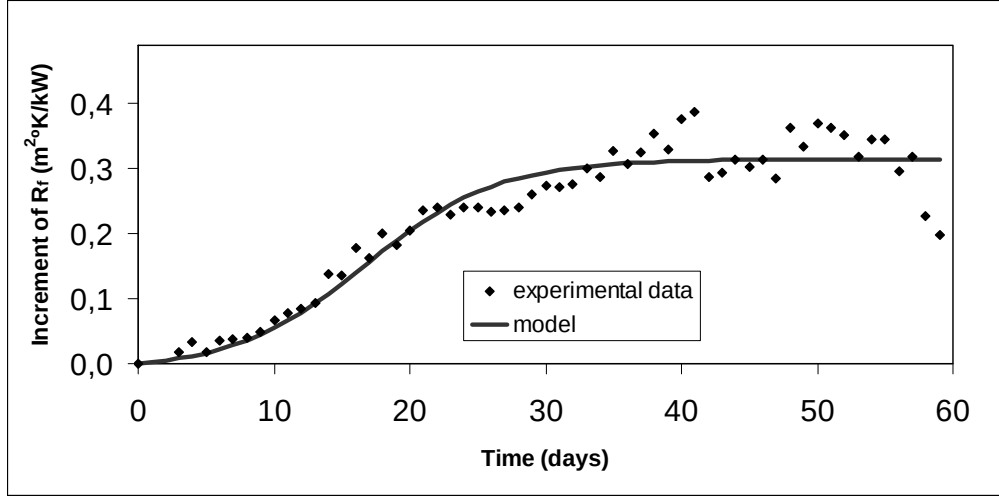


Figure 2.11. Example of proposed model and its fitting to experimental data.

After Konak (1973) model application to this experimental data, we could observe that it don't fit correctly to data. So, we decided to modify the model, but without increase the number of parameters. The new model is represented by equation 3. The model proposed in this paper to represent fouling evolution versus time, is a modification of the model proposed by Konak (1973).

$$\frac{dR_f}{dt} = k(R_{f\infty} - R_f) \cdot R_f \quad (3)$$

This model is just a combination of the first order equation and the driving force concept employed by Konak (1973), and it is a new expression of the logistic equation proposed by Verhulst in 1975 to interpret the biological population growths data. The model has the advantage that both kinetic parameters: $R_{f\infty}$ and k , presents clear physical significance. By one hand, $R_{f\infty}$ is the higher limit of the thermal resistance, and by the other hand k represents the rate which that maximum value is reached.

It can be seen in Figure 2.11 that this model fit correctly to data.

By integration of equation 3, it is obtained:

$$\ln \frac{R_f [R_{f0} - R_{f\infty}]}{R_{f0} [R_f - R_{f\infty}]} = k \cdot R_{f\infty} \cdot t \quad (4)$$

It is possible from equation 4, getting the variable R_f in function of the kinetics parameters k , $R_{f\infty}$ and R_{f0} (initial value of R_f), and time (equation 5). This fact permit to obtain easily by non linear regression both parameters, $R_{f\infty}$ and k .

$$R_{fi} = \frac{R_{f\infty}}{1 + \left[\frac{R_{f\infty}}{R_{fo}} - 1 \right] e^{-k \cdot R_{f\infty} \cdot t}} \quad (5)$$

Besides, it is of great concerning to get the value of $\tau_{1/2}$, time necessary for the growing of the half of fouling layer because it has a clear physical significance (it provide information about the time for fouling formation) and it is a useful parameter in order to designing and operating a heat transfer equipment. This time coincide with the inflexion point of the s-shape curve.

$$\tau_{1/2} = \frac{\ln \left[\frac{R_{f\infty}}{R_{fo}} + 1 \right]}{k \cdot R_{f\infty}} \quad (6)$$

3. A PARTICULAR *IN SITU* STUDY: “LOS BARRIOS” POWER STATION (ALGECIRAS BAY, SPAIN)

3.1. Assays Description

In this chapter are exposed the results of the study of different biocides effect on the formed biofouling inside of the tubes, from a scale pilot heat exchanger, that works under the same operation conditions that the heat exchanger of the Thermal Power station Los Barrios, Algeciras bay, South of Spain.

The fouling formed inside the tubes was characterized at physical-chemical level, using the methodology previously described. The same samples were also used to characterize the biofouling bacterial community deposited, untreated and treated with three different antifouling strategies, by the analysis of 16S rRNA gene using molecular techniques.

The biocides evaluated were an oxidizing compound, chlorine, a non-oxidizing product consists of a mixture of aliphatic amines, Mexel®432 and irradiation by UV light. The dosage regimen of each one is shown in the Table 3.6.

The experiences were carried out during a period corresponding to two month owing to the fact that previous assays in pilot plant have reported is the optimal period for the development biofilm.

It has also been studied the environmental effects of each one of the assayed biocides, analyzing their toxicity in different organisms and their persistence in the environment. And, finally an overall matrix for antifouling treatment choice is proposed.

3.2. Biofouling Characterization

3.2.1. Results of Physico-Chemical Characterization of Fouling

Effect on the Amount of Fouling

The effect of each antifouling treatment on accumulated fouling could be observed in Figure 3.1 and Table 3.1. In order to define a reference, in each experience a control tube (without treatment) was carried out. The chlorination was the treatment that reduced more efficiently the formation of fouling, and Mexel®432 and UV radiation presented similar results.

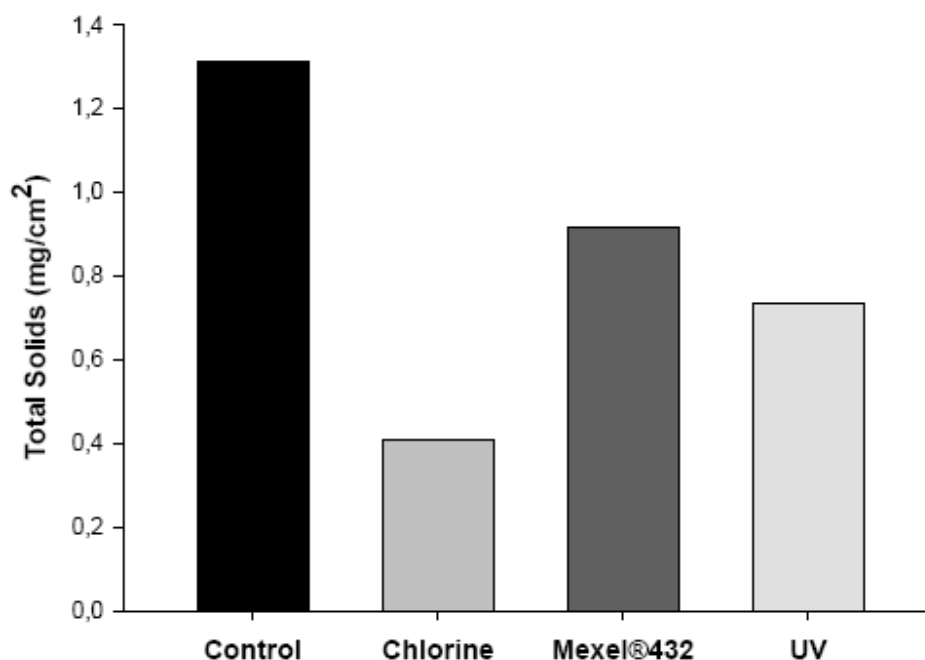


Figure 3.1. Concentration of fouling formed and percentage of amount reduction in relation with control tube.

Table 3.1. Wet Volume and thickness obtained for the 3 antifouling treatments and the control tube

Treatment	Wet volumen (cm ³)	Fouling thickness (μm)
Control	54	428
Chlorine	11	87
Mexel®432	33	261
UV	30	238

For all the treatments studied the wet volume and thickness of fouling were less than in the control tube, with minimum values when chlorine was used as antifouling product.

Effect on the Fouling Composition

The fouling formed under the particular site conditions was predominantly inorganic (Figure 3.2). Mexel®432 showed the higher percentage of organic matter due surely to its organic composition. The rest of tubes presented similar trend with respect to organic and inorganic matter (Table 3.2).

The data of elemental composition showed as fouling formed with Mexel®432, had a higher content in Carbon, in accord with the result from its higher organic proportion.

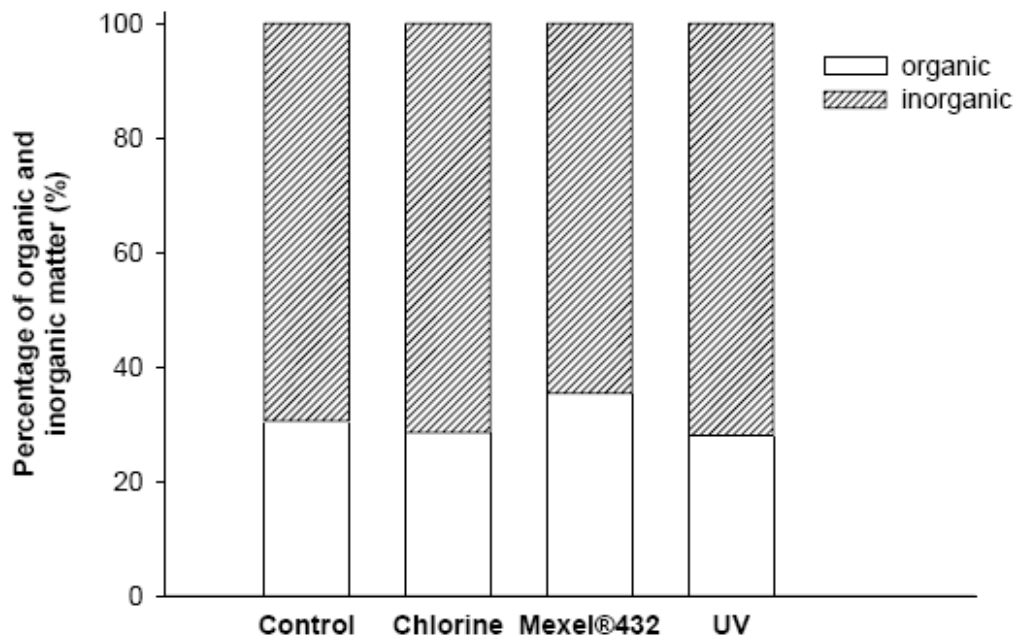


Figure 3.2. Organic and inorganic matter of fouling formed in the different treatment.

Table 3.2. Proportions of different elements in the fouling with different antifouling strategies

Treatment	C (%)	H (%)	N (%)	S (%)
Control	11.86	2.39	2.07	0.56
Chlorine	9.84	2.20	2.04	0.38
Mexel®432	15.22	2.89	2.55	0.56
UV	9.35	3.16	1.86	0.25

Effect on the Transport Properties

From data automatically monitored, controlled and recorded for each tube (temperatures, differential pressures, flow rates, etc.), it has been possible to follow the change of *heat transfer resistance* due to fouling (R_f) (Figure 3.3 and Table 3.3). In all the case, the values of R_f rises during the time of experiment due to an increase of biofilm.

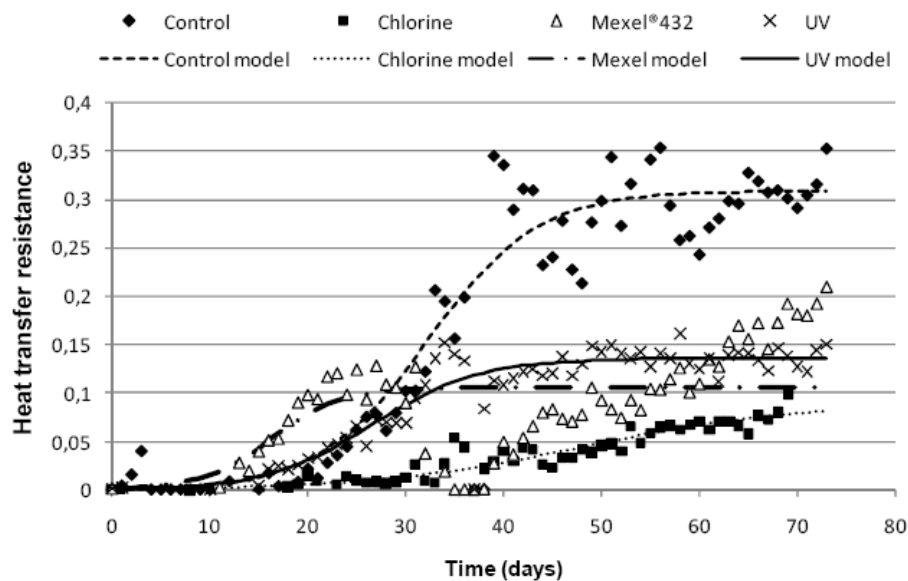


Figure 3.3. Experimental values and modelization of heat transfer resistance ($\text{m}^2 \cdot \text{K} \cdot \text{W}^{-1}$) of fouling formed using different treatment.

For an adequate interpretation of the R_f experimental data, was useful the application of the kinetic model proposed in section 2. It permitted to obtain the kinetic parameter: R_{fmax} , k and $t_{1/2}$ (Table 3.3). These parameters were registered, showing a clear decreasing of the maximal heat transfer resistance in the 3 tubes with antifouling treatment respect the control tube. In addition, chlorine was the treatment that delayed during more time the formation (greater $t_{1/2}$) of the fouling.

Table 3.3. Thermal conductivity (λ) and kinetic and statistical parameters of heat transfer resistance (R) obtained by the kinetic model

Treatment	R_{fmax} ($\text{m}^2 \cdot \text{K} \cdot \text{W}^{-1}$)	k	$\Sigma(\text{Pred-Obs})^2$	r^2	$t_{1/2}$ (days)	λ ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$)
Control	0.308	0.576	0.085	0.927	32.29	1.338
Chlorine	0.090	1.040	0.008	0.888	48.40	1.415
Mexel432®	0.106	2.679	0.141	0.586	16.41	2.541
UV	0.136	1.353	0.016	0.951	26.72	1.864

The estimation of the thermal conductivity of the fouling layer was the result of combination of direct (thickness) and indirect measurements (R_{fmax}), and provided valuable information about fouling behavior with respect to the heat transfer. In this way, it was observed that Mexel®432 though presented a high quantity of fouling, its elevated thermal conductivity did that it showed a R_{fmax} similar to chlorine. UV treatment presented the highest R_{fmax} value with respect to the other treatments, indicating that it wasn't a good alternative to chlorination as antifouling mitigation treatment.

3.2.2. Results of Molecular Characterization of Fouling

The fouling samples were analysed by molecular methods for bacteria identification. Altogether, 175 sequences were analyzed which corresponded to 83 different types of species or operative taxonomic units (OTUs). The rarefaction curve showed a clear plateau phase tendency, demonstrating that had been captured a sufficiently large proportion of the bacterial diversity present in heat exchanger tube (Figure 3.4). The results showed a clear dominance of *Proteobacteria* in biofouling deposited in our system.

The 16S rDNA gene libraries obtained from the biofilms revealed low bacterial diversity in the biofilm samples with chlorine and Mexel®432 in compare with UV light and control sample (Figure 3.5).

The *untreated biofilm* showed a high number of different phylogenetic groups, it sequenced clones related with (in increasing/decreasing order): *Proteobacteria*, *Bacteroidetes*, *Chloroflexi*, *Verrucomicrobia*, *Actinobacteria*, *Planctomycetes* and *Acidobacterias* indicating a complex bacterial community at 60 days of surface colonization. The sequences affiliated with the class γ -*proteobacteria* were the main group detected.

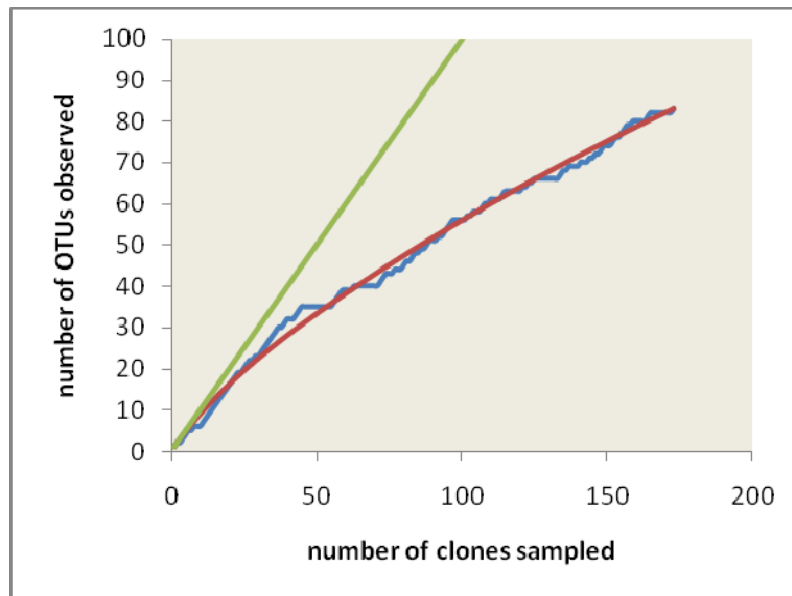


Figure 3.4. Rarefaction analysis for the biofilm community treated with biocides in heat exchanger tube from pilot plant.

The biofilm was dominated by filamentous γ -proteobacteria with *Thiothrix*-like and other sulfur-oxidizing groups just as methylotrophic bacteria affiliated with the class β -proteobacteria. Other filamentous bacteria were present in the untreated biofilm as members of *Cytophaga/Flavobacterium* of the *Bacteroidetes* phylum and other such as *Chloroflexi* and *Planctomycetes* showing the importance of filamentous bacteria in heat exchanger biofouling.

In the chlorinated biofilm the whole bacterial community from biofouling sampled belonged *Proteobacteria* phylum. α -proteobacteria was predominant phylogenetic group observed in the biofilm, suggesting that these organisms are well suited to survive in chlorinated water as reported Williams et al. (2003). These sequences were related mainly to the genera *Hyphomonas*, these bacteria produce an exopolysaccharide (EPS) capsule surrounds the entire cell which could play a role in biocide resistance.

In the biofilm treated with antifouling agent Mexel®432, γ -*Proteobacteria* clones dominated the library and were related to the genus *Marinobacter* and sulphur-oxidizing bacteria (SOB). The phylum *Bacteroidetes* constituted the second most important group.

In UV-treated system, five bacterial groups were obtained from the clone libraries, including the following divisions or phyla: *Proteobacteria*, *Bacteroidetes*, *Actinobacteria*, *Verrucomicrobia* and *Chloroflexi*. Four *Proteobacteria* subdivisions were present (α -, β -, δ -, and γ -*Proteobacteria*). The sequences affiliated with sulfur-oxidizing bacteria (SOB) of γ -*proteobacteria* subdivision were the dominant group in the sampled analyzed. In particular, clones related to *Thiothrix* spp were most frequently detected

In conclusion, our results show that different antifouling strategies produce changes on the bacterial community associated to biofilm. The rational use of complementary antifouling techniques, directed to control specific bacterial species, could be the most effective form in control of biofouling formation in heat exchangers.

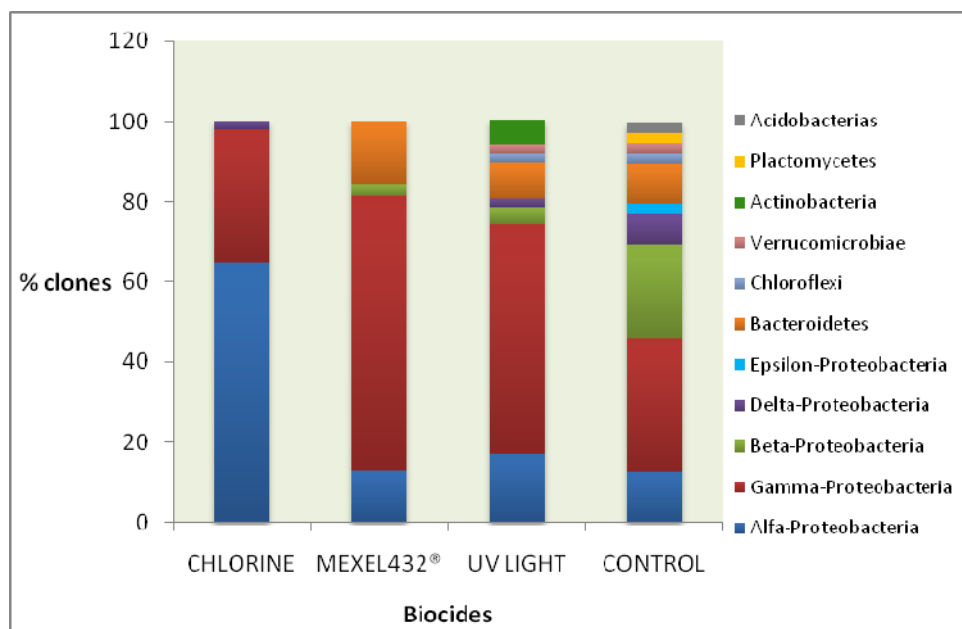


Figure 3.5. Percentage of clones in major bacterial lineages for biofilm 16S rDNA clone libraries.

3.3. Assessment Environmental of the Cooling Water Discharges

In this paragraph, we analyze the negative effect of the chemical antifouling, employed in the pilot plant, in marine organisms: chlorine and a non-oxidizing product consists of a mixture of aliphatic amines, Mexel®432.

3.3.1. Persistence of Chemicals in Seawater

The persistence tests were carried out in natural and filtered seawater and at controlled conditions of temperature (20 °C) and darkness. The primary degradation of aliphatic amine was tested following the guideline OPPTS (Office of Prevention, Pesticides and Toxic Substances) 835.3160 “Biodegradability in sea water”, proposed by the United States Environmental Protection Agency (USEPA, 1998). The concentrations of chemicals were periodically analyzed. Chlorine was analyzed by the DPD (*N,N*-diethyl-*p*-phenylenediamine) method (APHA et al., 1995). Mexel®432 was evaluated by methyl orange measurement after extraction with dichloro-1,2-ethane according to Crompton (1982).

Results were adjusted to different kinetics models. It was found that a logistic decay model led to better fitting, in both cases. At 0.3 mg/l, chlorine presented a rapid degradation in natural sea water, reached a decreased of 50% in 1 h and 90% in 3 h approximately (Figure 3.6). We have demonstrated that chlorine presented a rapid degradation in seawater.

On the other hand aliphatic amines showed more than 50% of primary biodegradation after 2 d and 90% after 15 d at an initial concentration of around 12 mg/l (Figure 3.7). This result indicated that, these Mexel®432 presented potential for slow but efficient degradation in seawater.

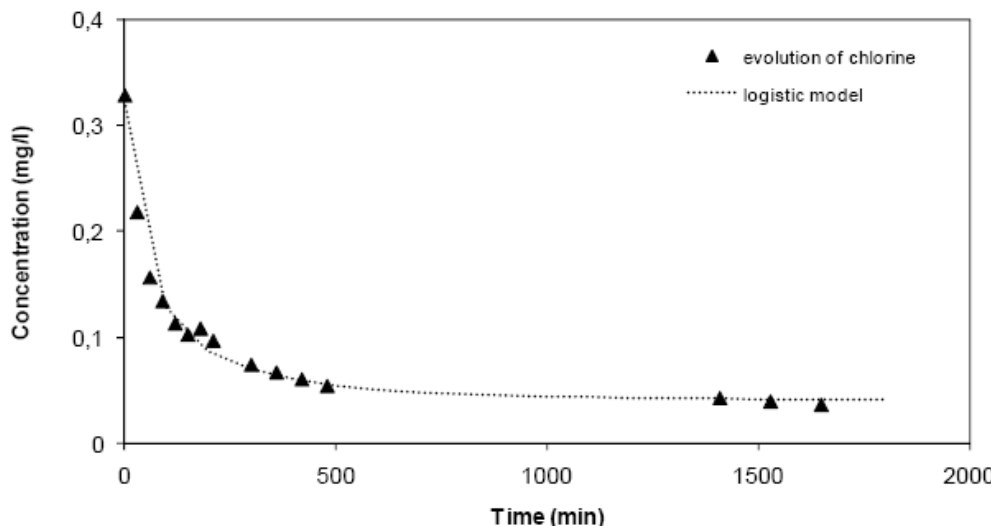


Figure 3.6. Evolution of chlorine in controlled conditions on seawater.

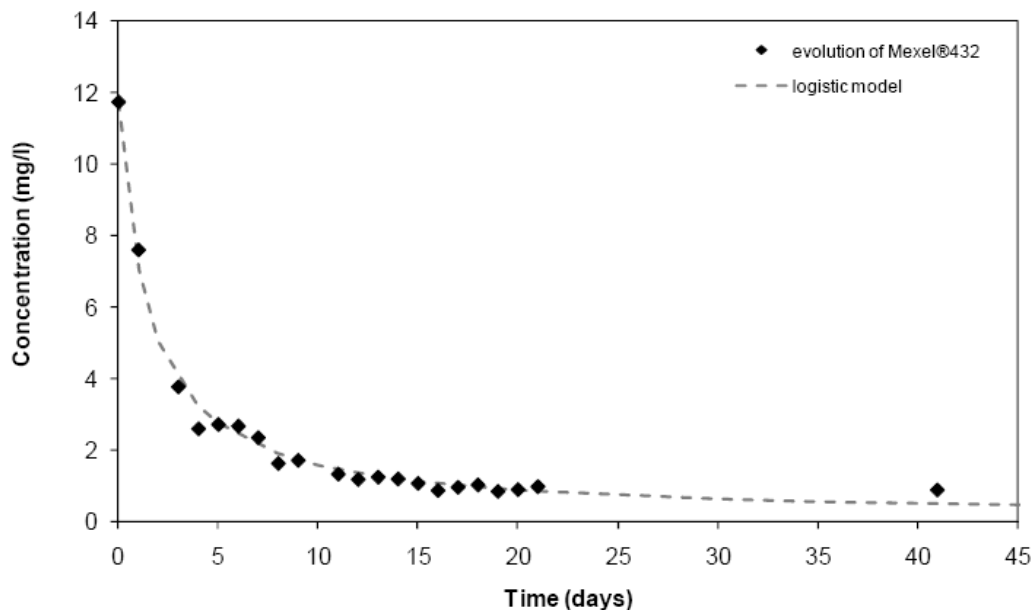


Figure 3.7. Evolution of Mexel®432 concentration in the biodegradation test.

3.3.2. Acute Toxicity of Chemical Antifouling

For microalgae toxicity, standardized tests in which the algae are cultivated and exposed to different concentrations of a toxic chemical under controlled conditions of light and temperature have been used (USEPA, 2002a, b). For the exposure period of 96-h, EC_{50} (and the 95% confidence range) values were estimate with the software IcPIN provided by the USEPA (Norberg-King, 1988) (Table 1). In addition, rotifer tests were conducted in sterile, 36-well polystyrene culture plates according with the protocol described by Snell and Perssone (1989). 24-h LC_{50} (lethal effect) for *Brachionus plicatilis* were analyzed by the PROBIT methods (USEPA, 2002a,b) (Table 3.4). We observed that chlorine showed a higher toxic effect in the marine organisms tested than Mexel®432.

Table 3.4. Effect concentrations (EC_{50}) of *Isochrysis galbana* and *Dunaliella salina* and lethal concentrations (LC_{50}) of *Brachionus plicatilis* exposed to different concentrations of chlorine and Mexel®432 after 96 and 24 h post-exposure, respectively

Chemical	<i>I. galbana</i>			<i>D. salina</i>			<i>B. plicatilis</i>		
	EC_{50} (mg/l)	Confidence limits lower	upper	EC_{50} (mg/l)	Confidence limits lower	upper	LC_{50} (mg/l)	Confidence limits lower	upper
Chlorine	2.91	2.79	3.00	1.73	1.62	1.85	1.23	1.16	1.31
Mexel®432	4.55	4.46	4.61	7.21	7.21	7.35	3.62	3.43	3.96

3.3.3. Chronic Toxicity of Chemical Antifouling

The chronic effect of chlorine and Mexel®432 at the concentration present in the discharge waters (0.2 mg/l of NaClO and 2.0 mg/l of Mexel®432) were evaluated in the flat fish *Solea senegalensis* and the common mussel *Mytilus galloprovincialis*. Different stress responses (as cortisol, glucose, lactate, Na^+/K^+ -ATPase [EC 3.6.1.3.], etc.) were assessed in plasmatic, branchial and renal tissue following the methodology described in Arjona et al. (2007). In addition, the xenobiotic metabolism responses were evaluated by means of enzymatic activities of ethoxyresorufin O-deethylase (EROD), glutathione S-transferase (GST, EC 2.5.1.18), glutathione reductase (GR, EC 1.6.4.2), glutathione peroxidase (GPX, EC 1.11.1.9), catalase (CAT, EC 1.11.1.6), carboxylesterase (CbE, EC 3.1.1.1.) and acetylcholinesterase (AChE, EC 3.1.1.7) in gill. Oxidative stress damage due to chemical exposure was measured by lipid peroxidation levels (LPO). The protocols of these enzymatic activities and the measure of LPO were described in more detail elsewhere (Solé et al., 2006; 2009).

In *Solea senegalensis*, chlorine induced mortality after two hours. Treating cooling seawater systems with chlorine (rather than with Mexel®432) seem to pose a greater threat to non-target biota. All the parameters measured differed from control values. Both antifoulings enhanced plasma cortisol levels and liver EROD activity during the first day of exposure. At the end of experiment, both products inhibited gill AChE activity.

In *Mytilus galloprovincialis*, our results showed that AChE, CAT and GST activities and LPO levels were sensitive indicators of exposure to these chemicals. Both tissues presented a differential response depending on the antifouling chemical used. Gill seemed a more adequate target tissue for evaluating chlorine exposure whereas digestive gland showed a more prominent role in the detoxification of Mexel®432.

3.4. Engineering and Economical Implications

Tree antifouling treatments are been compared: Electrochlorination, Mexel®432 and UV radiation. To evaluate the economic cost of antifouling treatments, we were going to take as example a coal power plant with the following operational and cost information.

Table 3.5. Initial data for economical study

General	Flow of cooling seawater	54 000 m ³ /h
	factor of emission of CO ₂ from coal combustion	0.879 kg/kWh.
	Cost of the electricity for industrial use	0.06 € for kWh
	Cost for CO ₂ emission (maximum price foreseen by Kyoto protocol)	40 €/Tn CO ₂ .
Electrochlorination	Performance of seawater electrolyzation	3.3 kWh/kg chlorine
	Active chlorine in the electrolyzers	1.0 g chlorine/l seawater.
UV disinfection	Lamp power consumption	150 W
	Lamp durability	12 000 h
	Number of lamps	1 024

Table 3.6. Economic study of the different treatments used

	Chlorine	Mexel®432	UV
Dosage	1.2 (mg/l)	5 (mg/l)	33 (mW s/cm ²)
Residual concentration	0.2 (mg/l)	2.5 (mg/l)	-
Time of application	continuously	30 (min/d)	continuously
Cost of product	198 (€/Tn)	5 400 (€/Tn)	400 (€/lamp)
Quantity of product	567.65 (Tn/year)	49 (Tn/year)	-
Electricity consumption (MWh/year)	1 873	minor	1 336
Operational cost (€/year)	112 394.30	266 085.00	80 732.16
Emission cost (€/year)	65863.06	minor	47 309.05
Annual reposition cost € (electrodes or lamps)	16 000	minor	299 008
Total cost (€/year)	194 257.37	266 085.00	427 049.21

The operational and reposition cost of the treatments, and the cost for the emission of CO₂ were showed in the Table 3.6. We observed that Mexel®432 and the chlorination presented a similar cost and they were smaller than the cost of UV treatment, that although have an electrical consumption lower than electrochlorination have a high cost for lamps reposition.

Technical Study of Antifouling Treatments

From a technical point of view, the facilities for Mexel®432 injection are very simple and no troubleshooting produces: only a tank and a small dosage pump are needed. In the other hand, electrochlorination need a relative complex installation and a very frequent maintenance. UV channels for as a great flow is not habitual, and many troubles can be predicted: lamp fouled or broken, electrical problems, etc.

4. CONCLUSION AND FUTURE PERSPECTIVES

4.1. Overall Evaluation Matrix for Antifouling Treatment Choice

To give a general and final vision of the result obtained and analyzed in the section 4 and 5 we used a matrix of comparison. In the Table 4.1, we gave a green color for the positive aspects, yellow for the neutral results and orange for the negatives, so that in ultimately we

could choose for the better treatment, considering engineering, economic and environmental aspects.

Table 4.1. Matrix of comparison of the different antifouling treatment study

Aspect	Chlorine	Mexel®432	UV
Engineering			
Economical			
Environmental			
Global valuation			

In conclusion, to evaluate new antifouling treatments we recommended follow the next studies for the different treatment:

1. Engineering
2. Economic

In addition, the following environmental test should be carried out for chemical antifouling:

3. Persistence or biodegradation
4. Toxicity

A more complex matrix must be developed including more detailed items with a relative weight according with its overall significance.

4.2. Microbial Methods to Control Biofouling: *Quorum Sensing*

In this chapter we introduce the concept of bacterial *quorum sensing* as a novel tool for biofouling control in industrial systems. This concept has already been applied successful in treatment as advanced wastewater treatment (Yeon et al., 2009). For many years bacteria were considered primarily as autonomous unicellular organisms with little capacity for collective behaviour. However, work in the last decade has shown that bacteria can communicate with each other by means of a process of communication named '*quorum sensing*'. It is population-density dependent and involves the production of signalling molecules that coordinate gene expression but that can diffuse between cells, hence affecting the whole population. *Quorum sensing* is responsible for modulating many different activities in diverse Gram-negative and Gram-positive bacteria including the expression of virulence factors, production of antibiotics and the formation of biofilms (Parsek and Greenberg, 2000; Chhabra et al., 2005).

Study have demonstrates an extended role of *quorum sensing* in biofilm formation and development. In particular, have determined that *quorum sensing* plays key roles in at least four stages of surface colonization in from attachment to swarming motility (Eberl et al., 1999), biofilm development (Labbate et al., 2004), and detachment (Rice et al., 2005) (Figure 4.1). Further, signalling molecules may also play other roles in biofilms, such as triggering the induction of a biofilm specific stress and biocide resistance (Van Houdt et al., 2004).

At present, the most effective methods of biofouling control are based on the application of chemical substances. All these chemicals are toxic and pollute the aquatic environment. Accordingly, there is a need for the development of ‘environmentally friendly’ nontoxic antifoulants. Because QS signals are important for bacterial growth, adhesion and biofilm formation, the inhibition of QS signals leads to the disruption of biofilm formation and, therefore, a way to control the marine biofouling could be inhibitors on *quorum sensing* by use QS blockers. One of them are molecules halogenated furanones produced by marine red algae, *Delissea pulchra*, which specifically inhibited AHL-regulated behaviours in a variety of Gram negative bacteria (De Nys et al., 1995; Yamazaki et al., 2006).

The first step in this approach is to investigate the capacity QS in bacterial isolated from natural biofilm formed in heat exchanger tube. A way is detecting the production of quorum signals in Gram-negative bacteria since practically whole bacterial community in our pilot plant belonging to this great group. In Gram-negative bacteria, the best investigated intercellular signaling compounds are the N-acylhomoserine lactones (AHLs) (Fuqua and Greenberg, 1998). The aim of this work was to detection of bacterial communication by means of N-Acyl Homoserine-Lactone Sensor Systems. We have been used two monitor system, *Chromobacterium violaceum* and *Escherichia coli* with pJBA130 plasmid fused green fluorescent protein (GFP). *Chromobacterium violaceum* characteristically produces violacein, a water-insoluble purple pigment. The production of violacein is regulated by the *quorum sensing* AHL molecule. CV026, is a mutant strain of the *C. violaceum* bacterium, cannot produce AHL, but by administering external AHLs to these bacteria, it can produce violacein.

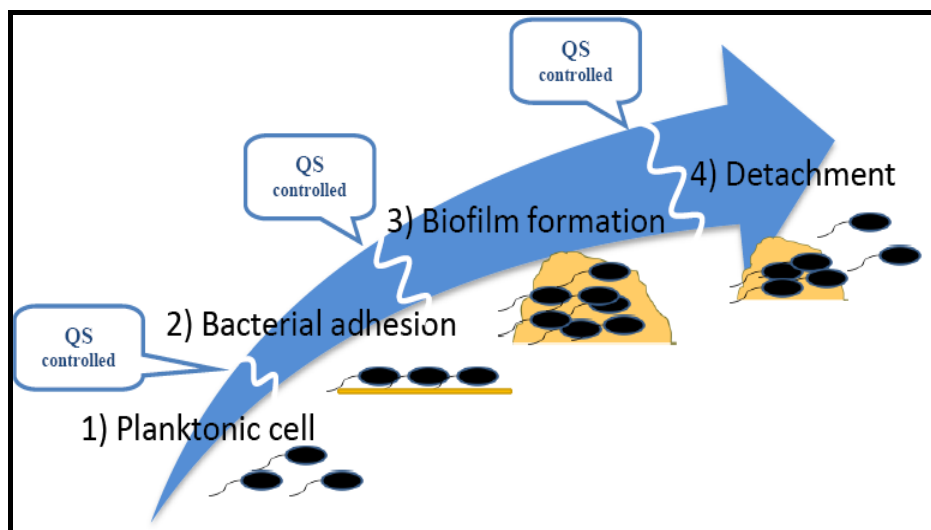


Figure 4.1. Model for quorum-sensing control of surface colonization.

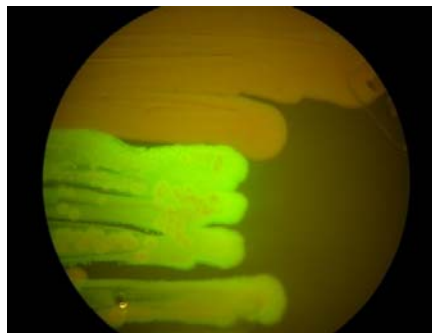


Figure 4.2. Induction of production green fluorescent in *E.coli* PJBA130 by synthetic AHLs.

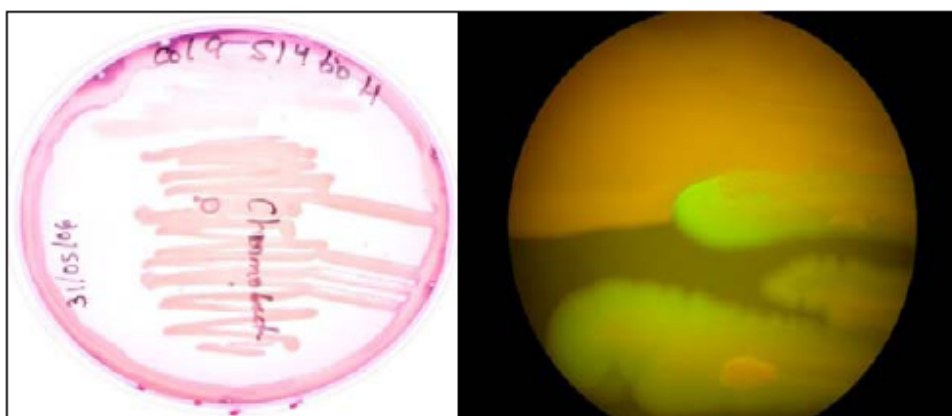


Figure 4.3. Presence of acylated homoserine lactones on bacteria isolated from heat exchanger biofilm. AHLs were determined by the sensor strain in *C.violaceum* CV026 and *E.coli* PJBA130.

Escherichia coli (pJBA130) acts AHL sensor that signals the presence of AHL molecules by expressing a reporter such as green fluorescent protein (GFP) (Figure 4.2)

To determine the capacity of these isolates to produce different types of cell density-dependent signalling molecules an agar-plate based bioassay was performed. Reporter strains, *C.violaceum* CV026 and *E.coli* (PJBA130), were cross-streaked against the isolates from heat exchanger tube biofilm to be tested on LB agar. *Serratia liquefaciens* served as positive control on both case.

In the presence of specific AHLs, CV026 will produce purple colonies due to violacein production and *E.coli* (PJBA130) will produce green fluorescent visualized in epifluorescence microscopy.

We report here, for the first time, that bacteria associated with biofilm from heat exchanger tube produce communication signals involved in *quorum sensing* in gram-negative bacteria.

One of 96 marine microorganisms isolated from natural biofilm was found to produce acylated homoserine lactones (AHLs) in assays based on the *Chromobacterium violaceum* CV026 and *E.coli* PJBA130 reporter system. The strain was identified by 16S ribosomal DNA gene sequence analysis as *Vibrio fisheri* (Figure 4.3).

It is likely that AHLs in other marine bacteria govern phenotypic traits as biofilm formation and biocide resistance in biofouling community.

Actually, it has been reported the production of AHL by *Roseobacter* strains isolated from marine snow (Gram et al., 2002). Therefore, it is expected that AHLs in *Roseobacter* species manage the formation of marine biofouling in heat exchanger since this genus has been detected in the most of samples of biofouling recollected from our pilot plant in early stages on biofilm formation. In the future, control strategies based *quorum sensing* (quorum quenching, QQ) suggests an interesting tool for biofouling control.

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